

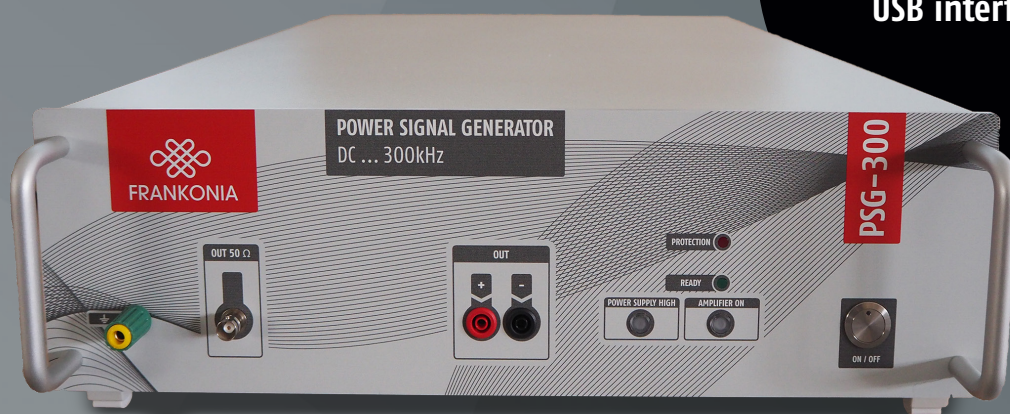


FRANKONIA
EMC Test-Systems GmbH

PSG-300

DATA SHEET

Software-based remote
control via the
USB interface.



Precision Generator

- ✓ For EMC tests IEC / EN 61000-4-16, IEC / EN 61000-4-19, IEC / EN 61543, IEC 60255
- ✓ Frequency range from DC to 300 kHz
- ✓ Available as: 5 A / 260 W (PSG-300)
16 A / 800 W (PSG-300A)
- ✓ Option: External power supply
Control input for an external voltage source (50 Ω), suitable for short tests up to 300 V. Can replace PSG-E300 with an added voltage source.
- ✓ Function generator for DC, sine, triangle, and square signals

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DESCRIPTION

The PSG-300 is an ultra-wideband linear power amplifier developed for signal frequencies from DC to 300 kHz. It is particularly suitable for demanding applications involving dynamic high-power signal generation. The integrated output stage provides up to 250 W (optionally 800 W) and can supply output currents of up to 5 A (16 A), featuring a fixed gain factor of 10.

A built-in waveform generator allows the creation of sine, square, and triangle signals, which are amplified internally by the power stage. External input signals can also be fed into the system through a separate input connection.

All device parameters and functions can be accessed through the supplied control software, allowing convenient remote operation over USB. Thanks to the available interface command set, the PSG-300 can also be integrated into automated measurement and test systems without difficulty.



TEST SETUP

TECHN. SPECIFICATIONS	AMPLIFIER	
	PSG-300 (5 A)	PSG-300A (16 A)
Frequency range	DC – 1 MHz (small signal –3 dB)	DC – 1 MHz (small signal –3 dB)
Performance range	DC – 300 kHz	DC – 300 kHz
Slew rate	100 V/μs	100 V/μs
Offset	± 1 mV (± 0.1 mV / °C)	± 1 mV (± 0.1 mV / °C)
Voltage amplification Option: ultra stable gain	10 ± 0.1 % (± 0.01 % / °C) 10 ± 0.1 % (± 10 ppm/°C)	10 ± 0.1 % (± 0.01 % / °C) 10 ± 0.1 % (± 10 ppm/°C)
Output voltage	50 Vrms / ± 75 Vpeak	50 Vrms / ± 75 Vpeak
Output current	5 Arms / ± 7.5 Apeak	16 Arms / ± 24 Apeak
Distortion (DC – 100 kHz, load ≥ 4 Ω)	< 0.10 %	< 0.10 %
Input impedance	100 kΩ	100 kΩ
Output impedance	<< 1 Ω and 50 Ω	<< 1 Ω and 50 Ω
Max. permissible input voltage	80 V (cont.), 100 V (< 1 min)	80 V (cont.), 100 V (< 1 min)
Broadband noise (10 Hz – 1 MHz, input 50 Ω)	0.5 mVrms	0.5 mVrms
Output power	260 W	800 W
Mains voltage	230 VAC / 50 Hz	230 VAC / 50 Hz
Control port	USB	USB
TECHN. SPECIFICATIONS	GENERATOR	
Frequency range	DC, 0.05 Hz – 300 kHz; resolution 0.05 Hz	DC, 0.05 Hz – 300 kHz; resolution 0.05 Hz
Frequency precision	± 20 ppm	± 20 ppm
Signal types	sine, square, triangular	sine, square, triangular



TECHN. SPECIFICATIONS	MECHANICAL DATA	
	PSG-300 (5 A)	PSG-300A (16 A)
Dimensions (W x H x D)	449 x 133 x 436 mm (3 RU)	449 x 177 x 585.5 mm (4 RU)
Weight	appr. 24 kg	appr. 32 kg

INCLUDED IN DELIVERY	
PSG300/A signal generator	
Operating manual	
USB drive including control software	
USB-A to USB-B cable, 3 m	
Power supply cable, 2 m	
BNC to BNC connection cable, 1.5 m	
USB-A / USB-B connection cable, 3 m	
Factory calibration certificate traceable to ISO 17025	

OPTIONS: ISOLATING TRANSFORMERS (ACC. TO EN 61558)
IT-06: isolation transformer, 1380 V, with integrated shielding winding between primary and secondary side; primary input: 230 V / 50–60 Hz; secondary output: 230 V / 6 A; enclosed in an aluminium housing.
IT-16: isolation transformer, 3680 VA, featuring a shielding winding between primary and secondary circuit; primary side: 230 V / 50–60 Hz; secondary side: 230 V / 16 A; robust aluminium enclosure.
IT-20: Isolating transformer 4600 VA with shielding winding between PRI and SEC; PRI: 230 V / 50–60 Hz; SEC: 230 V / 20 A; aluminium housing



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